

JEE MAIN REVISION MASTER: ELECTROMAGNETIC WAVES

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 15 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	EM Wave Characteristics & Spectrum	14 PYQs	EM Spectrum
Standard	Displacement Current & Maxwell's Equations	1 PYQs	Maxwell

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
EM Wave Characteristics & Spectrum	$c = \frac{1}{\sqrt{\mu_0 \epsilonpsilon_0}} = \frac{E_0}{B_0}, \quad u_{\text{avg}} = \epsilonpsilon_0 E_{\text{rms}}^2 = \frac{B_{\text{rms}}^2}{\mu_0}, \quad I = u_{\text{avg}} c = \frac{E_0 B_0}{2 \mu_0}$
Displacement Current & Maxwell's Equations	$I_d = \epsilonpsilon_0 \frac{d\Phi_E}{dt}, \quad \oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I_c + \epsilonpsilon_0 \frac{d\Phi_E}{dt} \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 EM Wave Characteristics & Spectrum (High Yield — 14 PYQs)

Core Principle: Transverse nature of EM waves ($\vec{E} \perp \vec{B} \perp \vec{k}$), Poynting vector, energy density, and frequency bands of spectrum.

■■ **Caution / JEE Pitfall:** In an EM wave, the energy is distributed EQUALLY between electric and magnetic fields: $u_E = u_B$.

■ **Speed Trick:** Radiation pressure on perfect absorbing surface: $P = I/c$; on perfect reflecting surface: $P = 2I/c$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "What range of electromagnetic spectrum is considered as light?..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Match the following types of waves with their wavelength ranges $\begin{matrix} \text{Waves} \\ \text{Wavelength ranges} \end{matrix}$..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "The total energy carried by the light wave when it travels from a rarer to a non-reflecting and nonabsorbing medium..." [Ans: N/A]

3.2 Displacement Current & Maxwell's Equations (Standard Yield — 1 PYQs)

Core Principle: Maxwell's correction to Ampere's law, displacement current inside charging capacitor dielectric space.

■ **Caution / JEE Pitfall:** Inside the charging capacitor plates, conduction current is 0 and displacement current equals I_c ; in the wire, displacement current is 0.

■ **Speed Trick:** Continuity of current holds everywhere: $I_{\text{(conduction)}} = I_{\text{(displacement)}}$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Match the following Maxwell's equations: (The symbols used here have their usual meanings) List-I List-II (a) Gauss' law for electrostatics ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

What range of electromagnetic spectrum is considered as light?

- (A) 1 mm to 700 nm
- (B) 400 nm to 1 nm
- (C) 400 nm to 700 nm
- (D) 1 nm to 10^{-3} nm

Correct Answer: None

Solution Analysis: The range of the electromagnetic spectrum considered as visible light is approximately 400nm to 700nm. This is the part of the electromagnetic spectrum that the human eye can detect. So, the correct option is: Option C: 400nm to 700nm

Benchmark Problem 2: KCET 2026 (Q#2)

Match the following Maxwell's equations: (The symbols used here have their usual meanings) List-I List-II (a) Gauss' law for electrostatics (i) $\vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$ (b) Gauss' law for magnetism (ii) $\vec{B} \cdot d\vec{A} = 0$ (c) Faraday's law (iii) $\vec{E} \cdot d\vec{A} = 0$ (d) Ampere-Maxwell's law (iv) $\vec{E} \cdot d\vec{A} = -\frac{d\phi_B}{dt}$

- (A) a - i, b - iii, c - iv, d - ii
- (B) a - ii, b - iii, c - i, d - iv
- (C) a - i, b - ii, c - iii, d - iv
- (D) a - ii, b - iii, c - iv, d - i

Correct Answer: None

Solution Analysis: Gauss' law for electrostatics says that the total electric flux through a closed surface is equal to the charge enclosed divided by ϵ_0 . So, $\vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$. Thus, (a) matches with (i). Gauss' law for magnetism says that the net magnetic flux through any closed surface is zero because isolated magnetic poles do not exist. So, $\vec{B} \cdot d\vec{A} = 0$. Thus, (b) matches with (iii). Faraday's law says that a changing magnetic flux produces an electric field. Its integral form is $\vec{E} \cdot d\vec{A} = -\frac{d\phi_B}{dt}$. Thus, (c) matches with (iv).